Assignment 1 - Git & GitHub Fundamentals

Session 1 - Group B | Individual Lab Report

Full Name: Hiba Aoufouchi Student ID: 118611

Course Code & Section: CSC 3324 01 GitHub Username: hibaaoufouchi4-bot

Repository: git-session1-hibaoufouchi4-bot

1. Repository Setup

I initialized the project with **git init -b main**. The **.gitignore** excludes **local.env** and ordinary **.log** files. An exception keeps **important.log** tracked.

2. README & Commit History

The README includes my details, the project banner, a short description, technologies, a Git command example, and a link to official Git documentation. The current main branch contains nine meaningful commits, including the About page merge and the GitHub README edit.

3. Selective Staging

I changed **style.css** and **notes.txt** at the same time, staged and committed only **style.css**, and left **notes.txt** modified. I used **git diff --cached** and **git diff** to inspect staged and unstaged changes.

4. Branch & Merge

I created **feature/about**, added an About page, and committed it on that branch. I then switched back to **main** and merged the feature branch.

5. Remote Changes

A commit records a snapshot locally; push sends commits to GitHub. Fetch downloads remote history without changing working files, while pull fetches and integrates remote changes. I edited the README directly on GitHub. **git fetch origin** showed local main one commit behind, while SHA-256 checks confirmed README.md and index.html stayed unchanged. **git pull --ff-only origin main** fast-forwarded main and updated the README.

6. Fair Use of AI

I used AI to interpret the checklist, help draft the README and report, and review the Git workflow. I checked the ignore rules, tracked log, branch history, and staged and unstaged changes with Git commands.

7. Sources & Reflection

Source: https://git-scm.com/docs. The hardest concept was keeping staged and unstaged changes straight. If Git behaves unexpectedly, I would first check **git status**, then inspect **git diff** and **git diff --cached**.